IT Workshop II (Cloud Computing): Introduction and Python Basics

July 28, 2026

Course Page and Evaluation

Course page will be on google classroom.
Class code: aawpyfln
Course Evaluation:
Lab Assignments: 40%
Quizzes: 15% (Two quizzes: both will be considered)
Mid-sem: 20%
End-Sem: 25%

Syllabus

CS351 **IT Workshop II (Cloud Computing)** **1-0-3-5**

Introduction to Cloud Computing, Cloud Concepts & Technologies, Cloud Services & Platforms, Hadoop & MapReduce - Concepts, Cloud Application Design, Python for Cloud, Cloud Application Development in Python, Big Data Analytics, Multimedia Cloud, Cloud Security, Cloud Application Benchmarking & Tuning

Texts:

1) Cloud Computing: A Hands-On Approach by Arshdeep Bahga, Vijay Madisetti, 2013, Universities Press

References:

1) Cloud Computing Bible by Barrie Sosinsky, 2011, Willey India Pvt Ltd

What is Cloud Computing?

The U.S. National Institute of Standards and Technology (NIST) defines cloud computing as:

Definition:

Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.

Cloud Computing Landscape

Applications



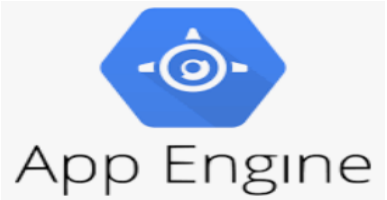
Storage



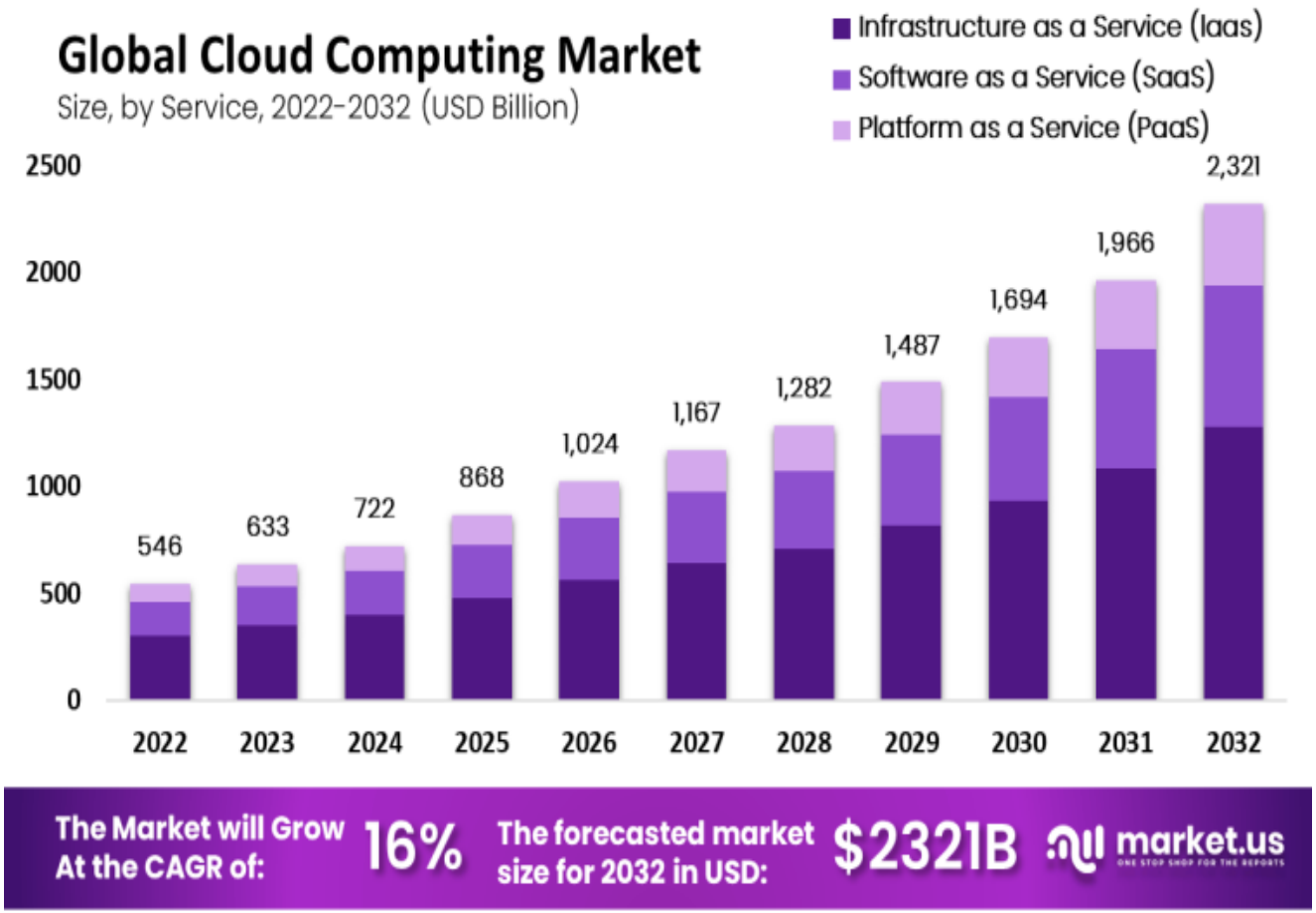
Computing



Platform



Cloud Computing Market

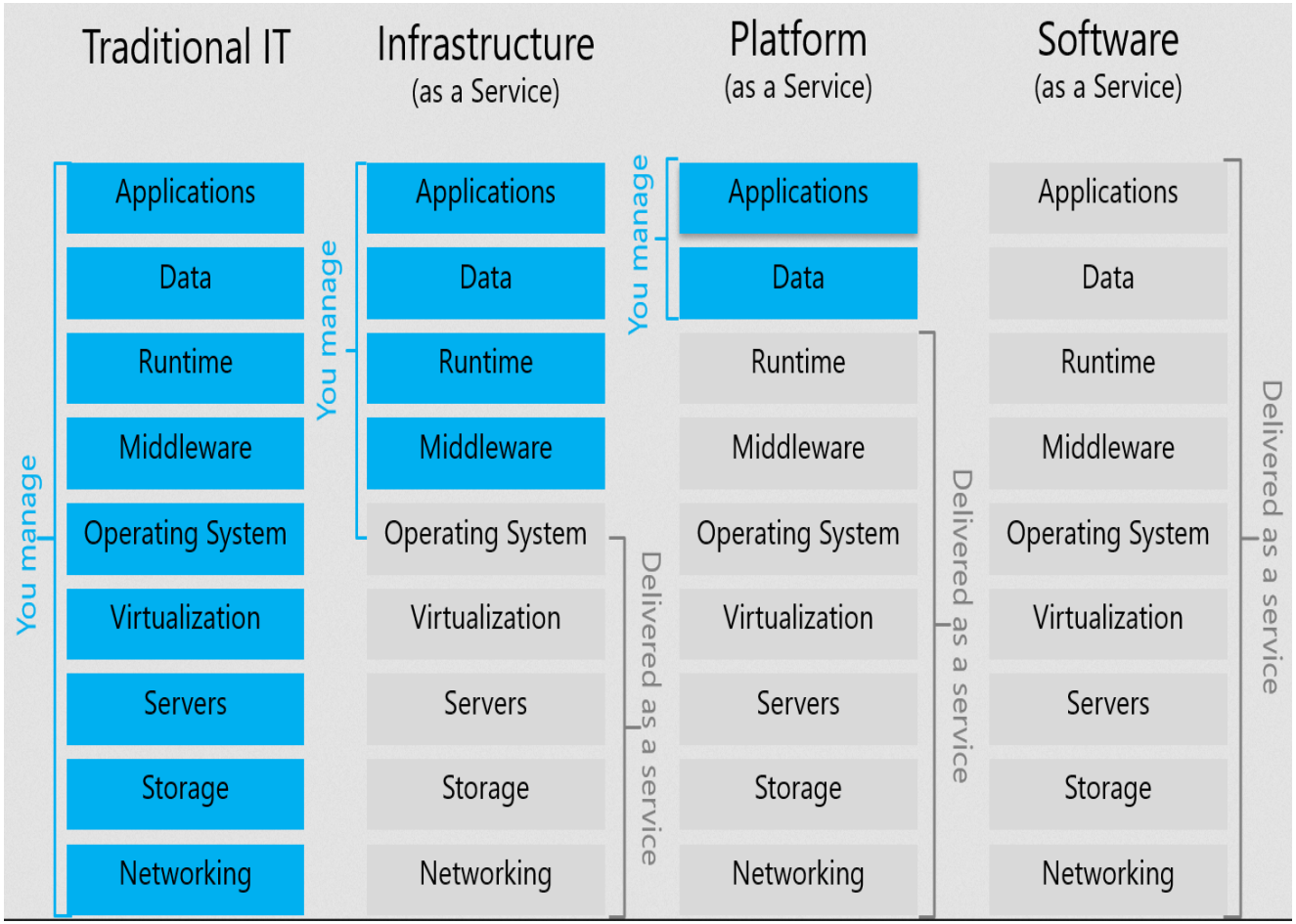


Source: www.techopedia.com

Characteristics of Cloud Computing

- On-demand self service
- Broad network access
- Multi-tenancy using virtualization
- Rapid elasticity - Horizontal scaling and Vertical scaling
- “Pay as you go” model
- Reduced costs
- Reliability

Cloud Computing Service Models



Source: Internet

The End